




Francesco Monastra

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Education

PhD in Astronomy, Astrophysics and Space Science

Sapienza University of Rome - University of Rome "Tor Vergata" - National Institute of Astrophysics, Italy

Enrolled: December 2025 (Ongoing)

Thesis: *Use of Artificial Intelligence for Particle Background Subtraction in X-ray Astronomy*

Supervisor: Dr. Simone Lotti (INAF-IAPS)

Co-Supervisors: Dr. Luigi Piro (INAF-IAPS), Prof. Elia Stefano Battistelli (Sapienza University of Rome).

M.Sc. in Astronomy and Astrophysics (LM-58)

Sapienza University of Rome, Italy

Enrolled: September 2022

Graduated: 26 May 2025

Grade: 110/110

Thesis: *Geant4 Simulations of Transition-Edge Sensors Detectors for the Large X-ray Observatory ATHENA and the Solar Axion Experiment IAXO*

Supervisors: Dr. Simone Lotti, Dr. Luciano Gottardi, Prof. Elia Stefano Battistelli

B.Sc. in Physics (L-30)

Sapienza University of Rome, Italy

Enrolled: September 2016

Graduated: 29 September 2022

Grade: 93/110

Thesis: *Acoustic Detection of High-Energy Astrophysical Neutrinos*

Supervisors: Prof. Irene Di Palma, Prof. Antonio Capone

Research Experience

Master's Thesis Internship at IAPS-INAF, Rome, Italy

April 2024 – May 2025

Supervisor: Dr. Simone Lotti

- Development and integration of CAD-based Mass Models for detailed Monte Carlo simulation;
- Study and implementation of particles induced environmental background for X-ray experiments;
- Definition of a pipeline for data analysis of simulations' results and implementation of the anti-coincidence operation between two detectors.

Master's Thesis Internship at SRON Netherlands Institute for Space Research, Leiden, Netherlands

October – December 2024

Supervisor: Dr. Luciano Gottardi

Funded by a dedicated scholarship of Fondazione Roma Sapienza (Prot.n. 61 del 19/03/2024, rep.n. 8/2024).

- Development and integration of CAD-based Mass Models for detailed Monte Carlo simulation;

- Development of a multi-step simulation with dedicated data analysis procedure;
- Characterization of environmental background: cosmic-ray showers, naturally occurring radioactive materials in laboratory environment and radionuclides in the cryostat layers;
- Definition of a pipeline for data analysis of simulations' results.

Astrophysics Laboratory Internship at IAPS-INAF, Rome, Italy

March – November 2023

Supervisor: Dr. Simone Lotti

Publications

Two papers in preparation as first author (provisory titles):

- **Reproducibility with Geant4 simulations of ground-based characterization campaign on TES-based Cryogenic AntiCoincidence Detector (CryoAC) of ATHENA X-IFU**
- **Background Geant4 simulation of transtion-edge sensor detectors and possible mitigation strategies for solar axion-like particles detection**

Two papers in preparation:

- **The ATHENA X-IFU Cryogenic AntiCoincidence detector (CryoAC): the new Demonstration Model (DM 1.1) and beyond**, *M. D'Andrea et al.*
- **Operation of the Cryogenic AntiCoincidence Detector (CryoAC) inside the ATHENA X-IFU Focal Plane Assembly Demonstration Model 1.1**, *M. D'Andrea et al.*

Technical Skills

Programming: C, C++, Python, Matlab, R

Simulation tools: Geant4, ROOT

3D Modeling tools: FreeCAD

Other: LaTeX, Git, Linux, Office Suite

Languages

Italian: Native

English: Advanced (C1)

Awards

Abroad Thesis Scholarship,

Fondazione Roma Sapienza, *Prot.n. 61 del 19/03/2024, rep.n. 8/2024*

Talks and Seminars

Background noise characterization on TES detectors with Geant4 simulations

SRON Leiden, Netherlands, 26 November 2024.

Geant4 Simulations of TES Detectors in the ATHENA and IAXO Experiments

INAF-IAPS Rome, Italy, 10 June 2025.

Academic Service and Outreach

Student Representative, Sapienza University of Rome

October 2018 – May 2025

Elected member in key university bodies including the **Department Council of Physics**, the **Faculty Board**, and the **University Didactic Commission**. Actively contributed to curriculum development, student policy discussions, and academic planning, advocating for student needs and ensuring transparency in university governance.

President, Knowledge Farm Association

2022 – 2024

Led the student association focused on science outreach and academic engagement at Sapienza.

Organizer and Moderator, Public Science Seminars

2020 – 2024

Developed and moderated events and seminars sponsored by Sapienza University, covering various physics and science topics for the general public.

Additional Experience

Private Tutor – Mathematics, Physics, Chemistry, Latin (2016 – Present)

Soft Skills

Teamwork, Public Speaking, Problem Solving, Leadership